



Date: 10th May, 2025

Timing: 2:00 P.M to 5:00 P.M

Max Marks: 100

Instructions:

1. All parts of a question should be answered together.
2. Make suitable assumptions whenever necessary with proper justifications.
3. Figures to the right indicate maximum marks assigned for that question.

Q.1. Answer the following questions. Each question carries 8 marks.

[48]

- (a) Design a advanced UML class diagram for an Online Food Delivery System with the following requirements. There are customers, restaurants, orders, and menu items. A customer can place multiple orders, each containing one or more food items. Each order is assigned to a delivery agent. A restaurant offers multiple food items and belongs to a category (e.g., Italian, Indian, etc.). Payment is associated with an order and can be via credit card, UPI, or cash. Identify different classes, attributes (e.g., orderId, price, deliveryAddress, etc.), and methods (e.g., calculateTotal(), trackOrder(), etc.). Show different relationships (e.g., association, aggregation, etc.) and multiplicities in the diagram.
- (b) Design a advanced UML state diagram for the Smart Traffic Management System. The system uses sensors and cameras to detect vehicle presence and traffic density in real-time at each direction (North, East, South, and West) of a traffic signal. It manages the traffic light cycle (Red, Green, Yellow) dynamically based on live data to reduce congestion and waiting time. Each traffic signal operates through a sequence of states. Models all possible states (Red, Yellow, Green, Emergency, Walking). Shows transitions triggered by (e.g., Timers (e.g., Green → Yellow after 30 sec), Sensor inputs (e.g., Red → Green if no cross-traffic), Emergency signals, etc.).
- (c) Design a detailed UML Sequence Diagram for a Smart Parking System. A Smart Parking System automates parking slot allocation, reservation, and payment process using IoT sensors and a mobile app. The system dynamically assigns slots based on vehicle type (Electric/Non-Electric), calculates fees, and processes payments. The system includes the following actors and components: Driver (User), Mobile app, Parking manager, Payment gateway, Parking slot, Vehicle, etc.,. It supports discount on elective vehicle. Also handles the failure conditions (e.g., payment fail, slot not available, slot booking fail, etc.).
- (d) Design a UML Activity Diagram for an Online Movie Ticket Booking System that captures the end-to-end workflow, from user login to ticket confirmation. The system should support login, multiple users, selects a movie, select location (cinema), date, time, real-time seat selection, payment processing, ticket generation, booking confirmation, and failure of any activity, etc.
- (e) A company's warehouse inventory system streamlines the item request and dispatch workflow between store managers, inventory management, databases, and logistics. You are required to design a UML Communication Diagram that captures this process by illustrating interactions between the following objects: Manager, InventorySystem, WarehouseDatabase, and LogisticsSystem.

- (f) Create a UML Use Case Diagram for a Online Video Streaming Platform where users manage their watchlists and playback preferences. Use cases include: (i) Watch a video. Stream content directly (ii) Add to watchlist (iii) Remove from watchlist (iv) Play next episode. Automatically triggered. (iv) Download for offline viewing. (v) Delete downloaded content. (vi) Rate a video. (vii) Comment on a video. (viii) View comments. (ix) Browse by genre. (x) Browse by actor/director. (xi) Create a user profile. (xii) Delete a user profile. Include the appropriate relationships for the use cases.

Q.2. Answer any Eight of following questions. Each question carries 5 marks.

[40]

- (a) Briefly describe the Booch system development processes.
- (b) Differentiate between correspondence, correctness, validation, and verification with respect to the object-oriented software development life cycle.
- (c) What are the key components of a UML Deployment Diagram? Describe the role of nodes, artifacts, and communication paths in a deployment diagram.
- (d) What is the role of UML Component Diagram in system design. Briefly describe the main elements of a UML Component Diagram, including components, interfaces, ports, and dependencies.
- (e) Discuss the role of packages in organizing a large-scale software system. How do they improve maintainability and scalability? Explain the various types of relationships that can exist between packages, such as import, access, merge dependencies.
- (f) Design black-box test cases for a login system with the following requirements: the username must be between 5 and 10 characters in length, and the password must contain at least 8 characters. Apply both boundary value analysis and equivalence partitioning techniques.
- (g) Perform statement coverage and branch coverage of white box testing on the following code. List the test cases.

```
def is_even(num):
    if num%2 == 0:
        return True
    else:
        return False
```

- (h) What is polymorphism in object oriented programming? Explain its different types with suitable examples.
- (i) Explain the potential risks or disadvantages of using inheritance in object-oriented programming with example.
- (j) What is coupling in object-oriented design? Explain the difference between tight coupling and loose coupling with examples.

Q.3. Define following terms with example: (i) Abstraction (ii) Encapsulation (iii) Qualifier (vii) Single Responsibility Principle

[12]

Best Wishes